



MS/MA IN ECONOMICS

DYNAMIC STOCHASTIC MODELS

Lecture 05

Linear Quadratic Method for Hansen's CIA Model

Tomás E. Baigorria

Martín A. Fradkin

Nicolás P. Monteverde Capello

Professor: George McCandless

July 29, 2026

Setup

We work with the cash-in-advance model, normalized by prices, exactly as presented in class, and log-linearize it around the stationary state.

The nonlinear model

Households solve

$$\max E_0 \sum_{t=0}^{\infty} \beta^t u(c_t^i, h_t^i), \quad u(c_t^i, h_t^i) = \ln c_t^i + \left[A \frac{\ln(1-h_0)}{h_0} \right] h_t^i,$$

subject to the budget constraint

$$c_t^i + k_{t+1}^i + \frac{m_t^i}{p_t} = w_t h_t^i + r_t k_t^i + (1-\delta)k_t^i + \frac{m_{t-1}^i + (g_t-1)M_{t-1}}{p_t},$$

and the cash-in-advance constraint (transfers usable in the period paid)

$$c_t^i = \frac{m_{t-1}^i}{p_t} + (g_t-1) \frac{M_{t-1}}{p_t}.$$

Substituting the CIA constraint into the budget constraint gives the simpler flow constraint

$$k_{t+1}^i + \frac{m_t^i}{p_t} = w_t h_t^i + r_t k_t^i + (1-\delta)k_t^i.$$

Production is $y_t = \lambda_t K_t^\theta H_t^{1-\theta}$, with technology following $\ln \lambda_{t+1} = \gamma \ln \lambda_t + \varepsilon_{t+1}$, and competitive factor markets give

$$w_t = (1-\theta)\lambda_t K_t^\theta H_t^{-\theta}, \quad r_t = \theta\lambda_t K_t^{\theta-1} H_t^{1-\theta}.$$

Aggregation requires $H_t = \int_0^1 h_t^i di$ and $K_t = \int_0^1 k_t^i di$.

Setting up the household's Lagrangian and taking first order conditions with respect to $c_t^i, h_t^i, k_{t+1}^i, m_t^i$, then combining them to eliminate the multipliers, delivers the two intertemporal conditions

$$\frac{1}{\beta} = E_t \frac{w_t}{w_{t+1}} [r_{t+1} + (1-\delta)], \quad \frac{B}{w_t} = -\beta E_t \frac{1}{c_{t+1}^i} \frac{p_t}{p_{t+1}}, \quad B \equiv A \frac{\ln(1-h_0)}{h_0}.$$

Aggregating across households ($C_t \equiv \int_0^1 c_t^i di$, and using the CIA constraint together with $M_t = g_t M_{t-1}$ at the aggregate level), the model in aggregate form – variables $\{K_{t+1}, Y_t, C_t, H_t, w_t, r_t, M_t, p_t\}$, shocks $\{\lambda_t, g_t\}$ – is the system of eight equations

$$1 = \beta E_t \frac{w_t}{w_{t+1}} [r_{t+1} + (1-\delta)], \tag{A}$$

$$\frac{B}{w_t} = -\beta E_t \frac{1}{C_{t+1}} \frac{p_t}{p_{t+1}}, \tag{B}$$

$$Y_t = \lambda_t K_t^\theta H_t^{1-\theta}, \tag{C}$$

$$w_t = (1-\theta) \frac{Y_t}{H_t}, \tag{D}$$

$$r_t = \theta \frac{Y_t}{K_t}, \tag{E}$$

$$C_t = g_t \frac{M_{t-1}}{p_t}, \tag{F}$$

$$M_t = g_t M_{t-1}, \tag{G}$$

$$K_{t+1} + \frac{M_t}{p_t} = w_t H_t + r_t K_t + (1-\delta)K_t. \tag{H}$$

Stationary state

Setting all variables constant, (A)-(H) reduce to

$$\frac{1}{\beta} = \bar{r} + (1 - \delta), \quad \frac{B}{\bar{w}} = -\beta \frac{1}{\bar{C}} \frac{1}{\bar{g}}, \quad \bar{Y} = \bar{K}^\theta \bar{H}^{1-\theta}, \quad \bar{w} = (1 - \theta) \left(\frac{\bar{K}}{\bar{H}} \right)^\theta, \quad \bar{r} = \theta \left(\frac{\bar{K}}{\bar{H}} \right)^{\theta-1},$$

$$\bar{C} = \bar{M}/\bar{p}, \quad \bar{M}/\bar{p} = \bar{w}\bar{H} + (\bar{r} - \delta)\bar{K}.$$

Log-linearization

Let a tilde denote the percentage deviation of a variable from its stationary-state value, e.g. $\tilde{K}_t \equiv (K_t - \bar{K})/\bar{K} \approx \ln(K_t/\bar{K})$, and let a bar denote the stationary-state value. Recall that the capital chosen at the end of period t (predetermined for $t+1$) is written $\tilde{K}_{t+1}$, following the convention that state variables dated one period ahead of t still belong to the “period- t ” block of the system.

Equations (C)-(G) are already in multiplicative (Cobb-Douglas or ratio) form, so each log-linearizes term by term. Taking logs of (C) and differentiating around the stationary state,

$$\ln Y_t = \ln \lambda_t + \theta \ln K_t + (1 - \theta) \ln H_t \implies 0 = \tilde{\lambda}_t + \theta \tilde{K}_t + (1 - \theta) \tilde{H}_t - \tilde{Y}_t. \quad (3)$$

The same operation applied to (D)-(G) gives

$$0 = \tilde{Y}_t - \tilde{H}_t - \tilde{w}_t, \quad 0 = \tilde{Y}_t - \tilde{K}_t - \tilde{r}_t, \quad (4)$$

$$0 = \tilde{g}_t + \tilde{M}_{t-1} - \tilde{p}_t - \tilde{C}_t, \quad 0 = \tilde{g}_t + \tilde{M}_{t-1} - \tilde{M}_t. \quad (6)$$

Equation (A) states that $(w_t/w_{t+1})[r_{t+1} + (1 - \delta)] = 1/\beta$ each period, in expectation. Taking logs of both sides and differentiating,

$$0 = \tilde{w}_t - E_t \tilde{w}_{t+1} + \frac{\bar{r}}{\bar{r} + (1 - \delta)} E_t \tilde{r}_{t+1} = \tilde{w}_t - E_t \tilde{w}_{t+1} + \beta \bar{r} E_t \tilde{r}_{t+1}, \quad (1)$$

where the stationary-state relation $\bar{r} + (1 - \delta) = 1/\beta$ was used to replace $1/[\bar{r} + (1 - \delta)]$ with β .

Equation (B) is $B/w_t = -\beta E_t(1/C_{t+1})(p_t/p_{t+1})$. Both sides are proportional to a product of level variables, so taking logs and differentiating gives

$$-\tilde{w}_t = -E_t \tilde{C}_{t+1} + \tilde{p}_t - E_t \tilde{p}_{t+1} \implies 0 = \tilde{w}_t - E_t \tilde{C}_{t+1} + \tilde{p}_t - E_t \tilde{p}_{t+1}. \quad (2)$$

Equation (H) is additive rather than multiplicative, so it is linearized directly: writing each level variable as (stationary state) $\times (1 + \text{tilde})$, e.g. $K_{t+1} \approx \bar{K}(1 + \tilde{K}_{t+1})$ and $M_t/p_t \approx \bar{M}/\bar{p}(1 + \tilde{M}_t - \tilde{p}_t)$, substituting into (H) and subtracting the stationary-state relation $\bar{M}/\bar{p} = \bar{w}\bar{H} + (\bar{r} - \delta)\bar{K}$ leaves only the first-order terms,

$$0 = \bar{w}\bar{H} \tilde{w}_t + \bar{w}\bar{H} \tilde{H}_t + \bar{r}\bar{K} \tilde{r}_t + [\bar{r} + (1 - \delta)] \bar{K} \tilde{K}_t - \bar{K} \tilde{K}_{t+1} - \bar{M}/\bar{p} \tilde{M}_t + \bar{M}/\bar{p} \tilde{p}_t. \quad (8)$$

Together with the two exogenous processes

$$\tilde{\lambda}_{t+1} = \gamma \tilde{\lambda}_t + \varepsilon_{t+1}^\lambda, \quad \tilde{g}_{t+1} = \pi \tilde{g}_t + \varepsilon_{t+1}^\phi, \quad (9)$$

this completes the log-linearization. Collecting equations (1)-(9), the full log-linearized system is

$$0 = \tilde{w}_t - E_t \tilde{w}_{t+1} + \beta \bar{r} E_t \tilde{r}_{t+1}, \quad (1)$$

$$0 = \tilde{w}_t - E_t \tilde{C}_{t+1} + \tilde{p}_t - E_t \tilde{p}_{t+1}, \quad (2)$$

$$0 = \tilde{\lambda}_t + \theta \tilde{K}_t + (1 - \theta) \tilde{H}_t - \tilde{Y}_t, \quad (3)$$

$$0 = \tilde{Y}_t - \tilde{H}_t - \tilde{w}_t, \quad (4)$$

$$0 = \tilde{Y}_t - \tilde{K}_t - \tilde{r}_t, \quad (5)$$

$$0 = \tilde{g}_t + \tilde{M}_{t-1} - \tilde{p}_t - \tilde{C}_t, \quad (6)$$

$$0 = \tilde{g}_t + \tilde{M}_{t-1} - \tilde{M}_t, \quad (7)$$

$$0 = \bar{w} \bar{H} \tilde{w}_t + \bar{w} \bar{H} \tilde{H}_t + \bar{r} \bar{K} \tilde{r}_t + [\bar{r} + (1 - \delta)] \bar{K} \tilde{K}_t - \bar{K} \tilde{K}_{t+1} - \bar{M}/\bar{p} \tilde{M}_t + \bar{M}/\bar{p} \tilde{p}_t, \quad (8)$$

together with the two stochastic processes

$$\tilde{\lambda}_{t+1} = \gamma \tilde{\lambda}_t + \varepsilon_{t+1}^\lambda, \quad \tilde{g}_{t+1} = \pi \tilde{g}_t + \varepsilon_{t+1}^\phi. \quad (9)$$

This is exactly the log-linearized system used in class, and from here we proceed as before.

Part (a): separating equations by whether they contain expectations

The instruction is to sort the eight equations according to whether they contain an expected (period $t + 1$) term. Equations (1) and (2) are the only ones with $E_t(\cdot)_{t+1}$ operators, so they go into the “second block”; equations (3)–(8) are purely contemporaneous or backward-looking and go into the “first block.”

For the first block we need to decide which variables are predetermined (entering the block with a one-period lag) and which are jump variables. Given equations (3)–(8), the only variables that appear lagged are $\tilde{K}_t$ (i.e. $\tilde{K}_{t+1}$ evaluated at $t - 1$) and $\tilde{M}_{t-1}$. Hence the state vector is

$$x_t = \begin{pmatrix} \tilde{K}_{t+1} \\ \tilde{M}_t \end{pmatrix},$$

and everything else – including $\tilde{p}_t$, since it has not been moved into the state vector at this stage – is a jump variable,

$$y_t = \begin{pmatrix} \tilde{Y}_t \\ \tilde{C}_t \\ \tilde{H}_t \\ \tilde{r}_t \\ \tilde{w}_t \\ \tilde{p}_t \end{pmatrix}, \quad z_t = \begin{pmatrix} \tilde{\lambda}_t \\ \tilde{g}_t \end{pmatrix}.$$

Writing (3)–(8) as $0 = Ax_t + Bx_{t-1} + Cy_t + Dz_t$, with variables ordered as above, gives

$$A = \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & -1 \\ -\bar{K} & -\bar{M}/\bar{p} \end{pmatrix}, \quad B = \begin{pmatrix} \theta & 0 \\ 0 & 0 \\ -1 & 0 \\ 0 & 1 \\ 0 & 1 \\ [\bar{r} + (1 - \delta)]\bar{K} & 0 \end{pmatrix}, \quad (10)$$

$$C = \begin{pmatrix} -1 & 0 & 1 - \theta & 0 & 0 & 0 \\ 1 & 0 & -1 & 0 & -1 & 0 \\ 1 & 0 & 0 & -1 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \bar{w} \bar{H} & \bar{r} \bar{K} & \bar{w} \bar{H} & \bar{M}/\bar{p} \end{pmatrix}, \quad D = \begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 1 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}. \quad (11)$$

Row five of C – the one coming from equation (7), $0 = \tilde{g}_t + \tilde{M}_{t-1} - \tilde{M}_t$ – is identically zero, because that equation contains no jump variable at all. A matrix with an all-zero row cannot have full rank, so $\det C = 0$ and C is singular. This is exactly the obstacle flagged in class: equation (7) only relates $\tilde{M}_t$ (a state variable) to $\tilde{M}_{t-1}$ and $\tilde{g}_t$, with no jump variable appearing, so it can never contribute a nonzero entry to C no matter how the remaining variables are ordered. Uhlig's method requires inverting C , so the block decomposition above cannot be carried through as written; this is precisely why the version worked out in class moved $\tilde{p}_t$ into the state vector instead of leaving it as a jump variable.

For completeness, the second block, built from equations (1)-(2) with the same x_t, y_t, z_t , is

$$0 = FE_t x_{t+1} + Gx_t + Hx_{t-1} + JE_t y_{t+1} + Ky_t + LE_t z_{t+1} + Mz_t,$$

with

$$F = G = H = L = M = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}, \quad J = \begin{pmatrix} 0 & 0 & 0 & \beta\bar{r} & -1 & 0 \\ 0 & -1 & 0 & 0 & 0 & -1 \end{pmatrix}, \quad K = \begin{pmatrix} 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \end{pmatrix},$$

since neither (1) nor (2) involves a state variable or a stochastic shock.

Part (b): collapsing everything into a single block

We now put every equation into the second block, so that there is no first block at all. Concretely, all eight endogenous variables are stacked into a single vector, in the order

$$x_t = \begin{pmatrix} \tilde{K}_{t+1} \\ \tilde{M}_t \\ \tilde{Y}_t \\ \tilde{C}_t \\ \tilde{H}_t \\ \tilde{r}_t \\ \tilde{w}_t \\ \tilde{p}_t \end{pmatrix}, \quad z_t = \begin{pmatrix} \tilde{\lambda}_t \\ \tilde{g}_t \end{pmatrix}.$$

There is no separate jump-variable vector y_t , so the matrices J and K disappear, and there is no first block, so A, B, C, D disappear as well. Every one of the eight equations (1)-(8) is now written directly as a row of

$$0 = E_t [Fx_{t+1} + Gx_t + Hx_{t-1} + Lz_{t+1} + Mz_t], \quad (12)$$

with the shocks evolving as $z_t = Nz_{t-1} + \varepsilon_t$. Reading off the coefficients of each of the eight equations on x_{t+1} , x_t , x_{t-1} and z_t (none of the equations contains an expected shock, so $L = 0$) gives

$$F = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & \beta\bar{r} & -1 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \quad (13)$$

$$G = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & -1 & 0 & 1 - \theta & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & -1 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 & -1 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\bar{K} & -\bar{M}/p & 0 & 0 & \bar{w}\bar{H} & \bar{r}\bar{K} & \bar{w}\bar{H} & \bar{M}/p \end{pmatrix}, \quad (14)$$

$$H = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \theta & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ [\bar{r} + (1 - \delta)]\bar{K} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \quad (15)$$

$$L = 0_{8 \times 2}, \quad M = \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 1 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad N = \begin{pmatrix} \gamma & 0 \\ 0 & \pi \end{pmatrix}. \quad (16)$$

The row order in (13)-(16) follows equations (1)-(8) exactly, and the column order in F , G , H follows the x_t ordering above; e.g. row 1 of G has a 1 in the $\tilde{w}_t$ column because equation (1) is $0 = \tilde{w}_t - E_t \tilde{w}_{t+1} + \beta \bar{r} E_t \tilde{r}_{t+1}$, and the corresponding entries of F carry the coefficients on $E_t \tilde{w}_{t+1}$ and $E_t \tilde{r}_{t+1}$.

Since every equation now lives in the same block, Uhlig's theorem for this case looks for an 8×8 matrix P and matrices Q solving the laws of motion

$$x_t = Px_{t-1} + Qz_t,$$

directly from the quadratic matrix equation

$$0 = FP^2 + GP + H, \quad (17)$$

with Q recovered from

$$\text{vec}(Q) = -(N' \otimes F + I_2 \otimes (FP + G))^{-1} \text{vec}(LN + M). \quad (18)$$

No inversion of a C matrix is required at any stage, which is precisely why this reformulation sidesteps the singularity found in part (a): the information that used to sit in the (non-invertible) first block is now folded directly into F , G and H . The price of this convenience is that (17) is a matrix quadratic equation in an 8×8 unknown rather than in the 3×3 (or 2×2) state block used in class, which is considerably more costly to solve by hand or numerically – consistent with why the class solution instead chose to enlarge the state vector by moving $\tilde{p}_t$ into it, restoring an invertible C with a much smaller P .